

Smart Campus Mobility: A Pilot Study of Ride-Sharing Adoption in Iraqi Universities

Research Proposal

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1. Background and Motivation

Urban transportation in Iraqi cities faces persistent challenges: chronic traffic congestion, limited public transit infrastructure, rising fuel costs, and a lack of integrated mobility solutions. University campuses in Iraq, which serve tens of thousands of commuting students daily, represent a microcosm of these broader problems. Most students rely on private vehicles, informal taxis, or inconsistent minibuses routes, resulting in long commute times, high costs, and significant environmental impact.

Ride-sharing platforms have transformed urban mobility worldwide. Empirical studies from the United States, Europe, China, and Southeast Asia have demonstrated that peer-to-peer ride-sharing can reduce vehicle kilometres traveled by 30–50%, lower commuting costs by 20–40%, and improve access to employment and education for underserved populations (Shaheen et al., 2016; Tirachini, 2020; Mouratidis et al., 2021). However, a critical gap persists in the literature: **there are zero peer-reviewed studies examining ride-sharing adoption in Iraq or comparable conflict-affected Middle Eastern contexts.** The socio-cultural factors that govern adoption in Iraq—including gender norms around mixed-gender travel, trust deficits in peer-to-peer transactions, cash-dominant economies, and security concerns—are fundamentally different from those studied in Western and East Asian settings.

Toosila is a fully developed, deployment-ready ride-sharing application specifically designed for Iraqi users. The platform incorporates culturally appropriate features such as gender-preference matching, family-member tracking, and integration with local payment norms. By piloting Toosila within the controlled environment of a university campus, this study aims to generate the first empirical evidence on ride-sharing adoption barriers and enablers in Iraq, contributing to both the transportation science literature and evidence-based mobility policy for Iraqi institutions.

2. Research Questions

This study addresses the following research questions:

RQ1

What are the primary barriers to ride-sharing adoption among Iraqi university students, and how do they differ from barriers reported in Western and East Asian contexts?

RQ2

How does interpersonal trust—both in co-riders and in the platform itself—evolve

over the course of a six-week ride-sharing pilot, and what factors predict trust formation?

RQ3

To what extent do gender, faculty affiliation, and commute distance moderate willingness to adopt ride-sharing, and do gender-preference matching features mitigate adoption resistance?

RQ4

What are the measurable cost and time savings achieved by active ride-sharing participants compared to their pre-pilot commuting behaviour and compared to a waitlist control group?

3. Methodology

3.1 Study Design

A quasi-experimental pre-post design with a waitlist control group will be employed. Participants will be randomly assigned to either the **treatment group** (immediate access to Toosila) or the **control group** (waitlisted for six weeks, then granted access). This design allows causal inference about the effects of ride-sharing access on commuting behaviour, cost, and attitudes.

3.2 Participant Recruitment

A minimum of **500 students** will be recruited from the Faculties of Engineering and Computer Science at the University of Kufa. Recruitment will proceed through:

- In-class announcements during regularly scheduled lectures (with instructor permission).
- QR-code posters placed in faculty buildings, cafeterias, and student common areas.
- Social media announcements on official university and departmental channels.

Eligibility criteria: (a) currently enrolled undergraduate or graduate student, (b) commutes to campus at least three days per week, (c) owns a smartphone capable of running the Toosila application. Participants will provide informed consent and may withdraw at any time without academic penalty.

3.3 Pre-Survey Instrument

Prior to the pilot period, all participants will complete a structured pre-survey capturing the constructs shown in Table 1.

Table 1: Pre-survey constructs and measurement scales.

Construct	Items	Scale
Current commute mode	Single choice (car, taxi, minibus, walking, other)	Categorical
Daily commute cost (IQD)	Open numeric	Continuous
Daily commute time (min, one-way)	Open numeric	Continuous
Willingness to share rides	5 items	7-point Likert
Interpersonal trust	Adapted from Gefen et al. (2003), 6 items	7-point Likert
Platform trust	Adapted from McKnight et al. (2002), 5 items	7-point Likert
Gender comfort in shared transport	4 items	7-point Likert
Technology readiness	Adapted from Parasuraman and Colby (2015), 8 items	7-point Likert

The survey will be administered in Arabic with back-translation validation.

3.4 Six-Week Pilot Period

Treatment-group participants will use the Toosila app for daily commuting over six weeks. The app will passively collect:

- Trip frequency, origin–destination pairs (anonymised to grid cells), and timestamps.
- Match rates (ride offers vs. ride requests fulfilled).
- In-app ratings and feedback.
- Feature usage logs (e.g., gender-preference filter activation, route deviation acceptance).

No GPS tracking data will be stored beyond anonymised grid-cell aggregation. Control-group participants will continue their normal commuting routines and complete weekly one-item commute-cost logs.

3.5 Post-Survey Instrument

After the six-week pilot, all participants will complete a post-survey including:

- Repeated measures from the pre-survey (commute cost, time, trust scales) to assess change.
- System Usability Scale (SUS) – 10 items (treatment group only).
- Net Promoter Score (NPS) – single item.
- Satisfaction with ride-sharing experience – 6 items, 7-point Likert.

- Open-ended: perceived barriers, suggestions for improvement.
- Willingness to continue using ride-sharing post-study – single item, 7-point Likert.

3.6 Analysis Plan

Quantitative analyses will include paired *t*-tests and mixed-effects models (group $\times$ time) for pre-post comparisons, structural equation modelling (SEM) for trust evolution pathways, and moderation analyses for gender and faculty effects. Qualitative responses will be analysed using thematic analysis (Braun and Clarke, 2006). Effect sizes (Cohen's *d*) and confidence intervals will be reported throughout.

4. Data Collection and Privacy

All study procedures will be submitted for approval to the University of Kufa Research Ethics Committee prior to data collection. The following safeguards will be implemented:

Anonymisation

All participant data will be de-identified upon collection. A unique study ID will replace names and student numbers. The linking key will be stored on an encrypted, air-gapped device accessible only to the PI.

Location privacy

GPS coordinates will be aggregated to 500 m grid cells before storage. Raw GPS traces will not be retained.

Data storage

Survey data will be stored on university-hosted servers. In-app behavioural data will be stored on encrypted cloud infrastructure with access restricted to the research team.

Informed consent

Participants will receive a bilingual (Arabic/English) consent form detailing data collection scope, storage duration (3 years post-publication), and their right to withdraw and request data deletion.

Compliance

The study will comply with Iraqi Ministry of Higher Education research guidelines and align with the principles of the Declaration of Helsinki.

5. Expected Outcomes

1. **First empirical dataset** on ride-sharing adoption in Iraq, filling a significant gap in the transportation literature for conflict-affected developing countries.
2. **Identification of adoption barriers** specific to the Iraqi context (trust, gender norms, security, payment preferences), enabling culturally informed platform design.
3. **Quantified cost and time savings** for ride-sharing participants, providing evidence for university administration to support campus mobility programmes.
4. **Trust evolution model** demonstrating how repeated positive interactions change trust toward peer-to-peer mobility in low-trust environments.
5. **Policy recommendations** for Iraqi universities considering subsidised or institutionally supported ride-sharing programmes.

6. Timeline

Table 2 summarises the study schedule. The total active study duration is 12 weeks; manuscript preparation extends to week 20.

Table 2: Study timeline.

Phase	Duration	Activities
Phase 1: Preparation	Weeks 1–3	Ethics approval, survey instrument validation, recruitment materials, app final testing
Phase 2: Recruitment	Weeks 4–5	Participant enrolment, consent collection, pre-survey administration, randomisation
Phase 3: Pilot Deployment	Weeks 6–11	Six-week active pilot (treatment group uses Toosila), weekly check-ins, technical support
Phase 4: Post-Survey	Week 12	Post-survey administration, data export, initial data cleaning
Phase 5: Analysis	Weeks 13–20	Statistical analysis, manuscript preparation, submission

7. Budget

Since the Toosila application is already fully developed and deployed, the budget requirements for this pilot are minimal (Table 3). No software development costs are required.

Table 3: Estimated budget.

Item	Estimated Cost (USD)
Cloud hosting and server costs (3 months)	150
Printed recruitment materials (posters, flyers)	100
Survey platform subscription (Arabic-compatible)	0
Participant incentives (mobile data top-ups for 500 students)	1,000
Research assistant (1 RA, part-time, 12 weeks)	600
Contingency (10%)	185
Total	2,035

8. Target Journals for Publication

The following journals are identified as suitable publication venues, listed by relevance and impact:

1. **Transportation Research Part A: Policy and Practice** (IF ≈ 6.3) – ideal for adoption and policy findings.
2. **Travel Behaviour and Society** (IF ≈ 5.2) – focus on behavioural aspects of ride-sharing in developing countries.
3. **Journal of Transport Geography** (IF ≈ 5.7) – spatial analysis of campus ride-sharing patterns.
4. **Sustainable Cities and Society** (IF ≈ 10.5) – sustainability and smart campus mobility angle.
5. **International Journal of Transportation Science and Technology** (IF ≈ 3.4) – applied transportation technology, open access.
6. **Case Studies on Transport Policy** (IF ≈ 2.8) – suitable for a focused campus pilot case study if a shorter paper is preferred.

A secondary publication targeting a regional conference (e.g., IEEE International Conference on Intelligent Transportation Systems) is also planned to disseminate preliminary results.

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